# Payload-Aware Two-Tier UAV Route Planning for Last-Mile Delivery

Review Report – I


Thirumurugan RA (3122235001149)
Meghna Manimaran (3122235001079)
Shrika SP Thota (3122235001125)

Guided by: Sujaudeen N, Assistant Professor

Department of Computer Science and Engineering
Sri Sivasubramaniya Nadar College of Engineering


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# Presentation Outline


1. Introduction
2. Literature Review
3. System Design
4. Implementation and Results
5. Project Management
6. Conclusion

# Background and Motivation

- UAVs are an effective platform for last-mile delivery where ground transport is impractical: remote clinics, emergency drug drops, and fragile diagnostic sample transport.
- Every flight must balance three **conflicting objectives**: total distance, battery energy use, and exposure to high-risk flight segments.
- Existing routing systems treat all deliveries **identically** — a fragile medical device and a robust parcel travel the *same* route.
- Payload metadata (fragility, urgency, payload weight) encodes routing priorities, but **no system exploits this at mission time**.

**Core Motivation**

Payload-blind UAV routing pipelines cannot adapt to cargo-specific constraints. A two-tier architecture that bridges payload semantics directly to Pareto route selection addresses this structural gap.

# Real-World Cost of Payload-Blind Routing

## Swiss Post – Matternet M2 Crashes (Zürich, Jan & May 2019) [8]

Drones carrying clinical laboratory specimens flew **fixed, payload-invariant routes**. Post-incident analysis confirmed identical corridors regardless of cargo type — no routing adjustment for payload fragility.

## Vibration & Turbulence Degradation Studies [9, 10, 11]

Controlled lab work shows drone-transported medicines degrade measurably: insulin fibrillation, biologic protein denaturation, elevated haemolysis in blood samples on high-vibration segments. Altitude variation and turbulence raise lab rejection rates.

## Zipline Rwanda Blood-Delivery Network (up to 80 km/charge) [12]

Heavier blood-product payloads draw more hover/lift power and shorten range, yet the system flies the same fixed route regardless of payload weight — no energy-efficient detour for heavy cargo, no direct short route for urgent light payload.

**Root cause in every case:** the planner does not know what is carried, so it cannot prefer smoother corridors for fragile payloads or shortest paths for time-critical medicine.

# Problem Statement

## Problem

Existing UAV routing systems apply **static, cargo-agnostic objective weights**, resulting in **suboptimal or unsafe routes** for payloads with differing fragility, urgency, and payload weight.

- Given payload metadata $(F, W, U, P_{\text{lvl}})$ with $F \in \{1 \dots 5\}$, $W \in \{1 \dots 3\}$, $U \in \{1 \dots 3\}$, $P_{\text{lvl}} \in \{1 \dots 5\}$, and airspace graph $G = (V, E)$, select route $p^*$ from the Pareto front that best satisfies payload requirements — no expert re-tuning, no re-running the search.
- Single-objective routing collapses all objectives into one scalar and cannot model their three-way conflict.
- No existing system maps payload description to Pareto point selection via an explicit algebraic formula.
- Constraints: selection must be deterministic, reproducible, and finish within 1 s of payload tag entry.

# Response to Review 0 Panel Comments

| Panel Comment | Resolution in Current Work |
|---|---|
| **"No correlation between Tier 1 and Tier 2 factors"** | Explicit algebraic mapping introduced: Fragility $F$ scales $w_{\text{risk}}$; Urgency $U$ scales $w_{\text{dist}}$; Payload weight $W$ scales $w_{\text{energy}}$. Every Tier 2 tag maps to a Tier 1 objective via a closed formula. |
| **"Criteria are not considered to map together"** | The formula $w_{\text{risk}} = (F{\times}2)\,/\,(F{\times}2 + U{\times}3 + W{\times}2)$ (and equivalents) is a continuous function — all four tags simultaneously influence all three weights; changing any single tag shifts all three weights proportionally. |
| **"Application of Pareto needs rethinking"** | Extreme-point selection **replaced** by utility scoring over the *full* Pareto front: $\text{score}(p_i) = w_d(1-\tilde{d}_i) + w_\varepsilon(1-\tilde{\varepsilon}_i) + w_\rho(1-\tilde{\rho}_i)$. Mixed-priority payloads select *non-extreme interior points* that no single-objective baseline can produce. |
| **"Complexity only in Tier 2; more than 2 objectives"** | Tier 1 manages three-objective NAMOA* graph-search complexity; Tier 2 manages metadata-to-weight mapping and full-front utility scoring. Both tiers carry distinct computational roles — neither reduces to a single-objective problem. |

# Project Objectives

1. To build a **multi-objective A* engine (Tier 1)** that constructs a Pareto-optimal route set over an airspace graph, balancing distance, energy consumption, and flight risk simultaneously.
2. To design an **algebraic bridge** that translates payload metadata tags (fragility $F$, payload weight $W$, urgency $U$, priority $P_{\text{lvl}}$) into normalised routing objective weights with no expert tuning.
3. To implement a **Tier 2 payload-driven selector** that scores every Pareto route using the derived weights and returns the utility-maximising route $p^*$.
4. To evaluate the pipeline using **simulation benchmarks** on 50, 200, and 500-node graphs against single-objective A* and MOPSO baselines, using hypervolume, Pareto spread, IGD, and Tier 2 selection accuracy.
5. To validate the system on **real drone hardware** (Raspberry Pi 5 + Pixhawk) through ArduPilot SITL simulation and outdoor flight tests.

# Scope and Expected Outcomes

## Included

- Payload-aware UAV route selection for single-drone delivery
- Two-tier pipeline: Pareto front construction + utility-based Pareto point selection
- Software simulation on OSM graphs (Thiruvanmiyur, Chennai)
- ArduPilot SITL + RPi 5 + Pixhawk hardware integration and outdoor flight tests
- Benchmark against single-objective A* and MOPSO

## Excluded / Constraints

- Multi-UAV fleet management and coordination
- Real-time dynamic obstacle avoidance during flight
- Commercial deployment and regulatory compliance
- Weather-based in-flight re-routing
- Deep-learning-based risk estimation

# Literature Review

| Author / Year | Method | Key Contribution | Limitation / Research Gap |
|---|---|---|---|
| Chokkalingam & Ganesan (2026) [1] | Pareto-entropy PSO (POPSO) with dynamic parameter adaptation | Entropy-guided archiving improves front diversity; adaptable to real-time environmental changes | Final route still a manually re-tuned weighted sum per mission; payload metadata never incorporated — fragile samples and emergency medicine get the same hand-tuned selection |
| Tsoutsouras et al. (2020) [7] | Mass-aware flight-speed and energy-model adaptation for multi-drop trajectories | Reduces energy consumption vs. mass-agnostic planning across multi-stop routes | Continuous mass used only inside the energy/speed model itself; builds no multi-objective Pareto front; discrete payload metadata (fragility, urgency) plays no role in route selection |
| Ren et al. (2019) [2] | Crowding-distance NSGA-II (CDNSGA-II) over octree urban environment | Well-spread, collision-free Pareto-optimal paths in urban airspace | Crowding distance encodes only geometric front spread; urgency, fragility, and payload weight play no role in selection |

# Research Gap and Proposed Contribution

## Three Identified Gaps

- **Gap 1:** No payload-aware route *selection* — all planners apply a fixed geometric criterion independent of payload type, or no selection mechanism at all
- **Gap 2:** No front reuse — full multi-objective search re-run whenever mission context changes
- **Gap 3:** No algebraic bridge — no formalised mapping from payload description to the routing trade-off it implies

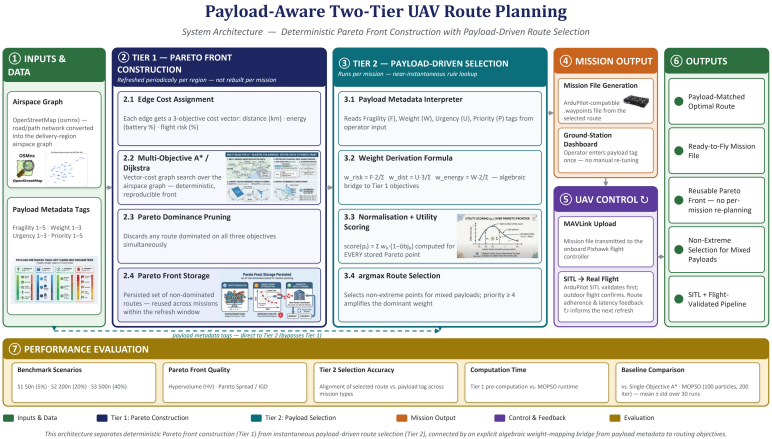
## Proposed Contribution (closes all three)

- Tier 2 maps four payload tags directly to Pareto point selection via formula **(Gap 1)**
- Front refreshed periodically; Tier 2 selects per mission against current front — no re-run **(Gap 2)**
- Closed-form bridge: $F, W, U \rightarrow w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}}$ **(Gap 3)**
- $p^* = \arg\max_{p_i \in \mathcal{P}} \text{score}(p_i)$ — selects non-extreme interior points for mixed payloads

# Existing System vs. Proposed System

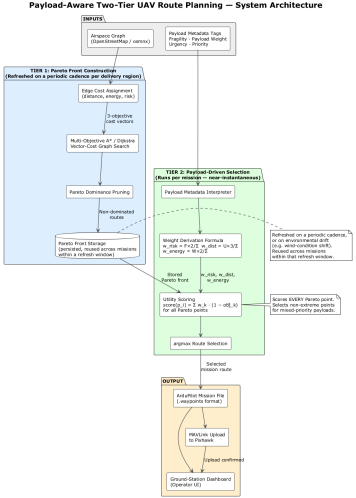
| **Criterion** | Existing Routing Systems | Proposed Two-Tier System |
|---|---|---|
| **Routing approach** | Single-objective (distance or energy only) | Multi-objective NAMOA* — distance, energy, and risk simultaneously |
| **Payload handling** | Cargo-agnostic; all parcels treated identically | Payload tags ($F, W, U, P_{\text{lvl}}$) derive objective weights algebraically |
| **Route selection** | Single greedy path or extreme Pareto point; expert re-tunes weights per mission | Utility scoring over full Pareto front; selects non-extreme interior points for mixed payloads |
| **Front reuse** | Full search re-run on every mission context change | Front built once per region; Tier 2 selects in $<1$ s against current front |
| **Hardware validation** | Simulation only | ArduPilot SITL + RPi 5 + Pixhawk outdoor flight tests |

# Proposed Methodology




Tier 1 Pareto front construction feeding Tier 2 payload-aware route selection. The front is built once per delivery region and reused across all missions within the refresh window.

# Proposed System Architecture



# System Modules

1. **M1 – Airspace Graph Builder:** Extracts OSM road/path network via `osmnx`; assigns three objective costs to each edge: geodesic distance, energy (distance $\times$ altitude-change factor), and flight risk (simulated threat zones and obstacle density, scored 0–1).
2. **M2 – Multi-Objective A* Engine (Tier 1):** NAMOA* search maintaining a cost vector $\mathbf{g} = [d, \varepsilon, \rho]$ per partial path; node-level label-dominance pruning; stores deterministic Pareto front $\mathcal{P}$ on disk for Tier 2 reuse.
3. **M3 – Payload Metadata Interpreter (Tier 2):** Reads tags $(F, W, U, P_{\text{lvl}})$; derives $w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}}$ via closed formula; min-max normalises objectives; returns arg max route and generates ArduPilot `.waypoints` mission file.
4. **M4 – Testing and Evaluation:** Benchmarks on S1/S2/S3 graphs vs. single-objective A* and MOPSO (30 runs); metrics: hypervolume, Pareto spread ($\Delta$), IGD, Tier 2 selection accuracy, computation time. Extends to ArduPilot SITL and outdoor flight tests.

# Methodology / Workflow

1. Literature review and problem formulation
2. Airspace graph construction (OpenStreetMap + OSMnx)
3. Multi-objective A* (NAMOA*) engine implementation (Tier 1)
4. Algebraic bridge and payload-driven Pareto selector (Tier 2)
5. Software simulation and benchmarking vs. A* and MOPSO baselines
6. Hardware integration (RPi 5 + Pixhawk), ArduPilot SITL simulation, and outdoor flight validation

## Evaluation Metrics

- Hypervolume (HV)
- Pareto spread ($\Delta$ metric)
- IGD vs. reference front
- Tier 2 selection accuracy
- Tier 1 computation time
- Route adherence (GPS error)

# Core Algorithm: Payload-Aware Two-Tier Route Planning

## Algorithm Summary

1. **Input:** Graph $G = (V, E)$, payload tags $(F, W, U, P_{\text{lvl}})$, source $s$, destination $d$.
2. **Weight derivation (Algebraic Bridge):**
   $r_{\text{raw}}{=}F{\times}2, \quad d_{\text{raw}}{=}U{\times}3, \quad e_{\text{raw}}{=}W{\times}2; \quad$ normalise $\Rightarrow w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}}$.
   If $P_{\text{lvl}} \geq 4$: amplify dominant weight by $0.1{\times}w_{\max}$, then renormalise to sum to 1.
3. **Tier 1 (NAMOA*):** Run multi-objective A*; maintain cost vectors; apply node-level label-dominance pruning; store Pareto front $\mathcal{P}$ on disk for reuse.
4. **Tier 2 — Utility Scoring:** For each $p_i \in \mathcal{P}$,

$$\text{score}(p_i) = w_{\text{dist}}(1 - \tilde{d}_i) + w_{\text{energy}}(1 - \tilde{\varepsilon}_i) + w_{\text{risk}}(1 - \tilde{\rho}_i)$$

5. **Output:** $p^* = \arg\max_{p_i \in \mathcal{P}} \text{score}(p_i)$; generate ArduPilot `.waypoints` file; upload via MAVLink.

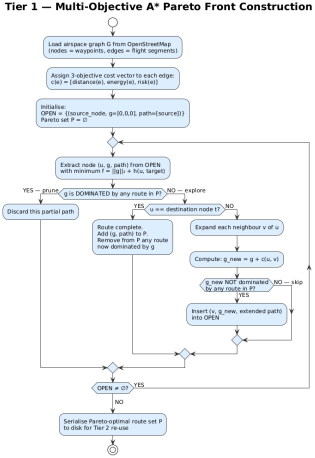
# Technology Stack and Resources

| Category | Selected Technology / Resource |
|---|---|
| **Programming language** | Python 3.11 |
| **Core libraries** | NetworkX, OSMnx, NumPy, matplotlib, SciPy |
| **Communication protocol** | MAVLink / ArduPilot `.waypoints` mission format |
| **Simulation environment** | ArduPilot SITL (primary validation before real flights) |
| **Map / dataset** | OpenStreetMap airspace graph (Thiruvanmiyur, Chennai urban area) |
| **Development tools** | Git, VS Code, QGIS, Mission Planner, ArduPilot SITL, Docker |
| **Hardware** | Raspberry Pi 5 (quad-core 2.4 GHz, 8 GB), Pixhawk 6C, GPS module, F450 frame, ESC + motors |

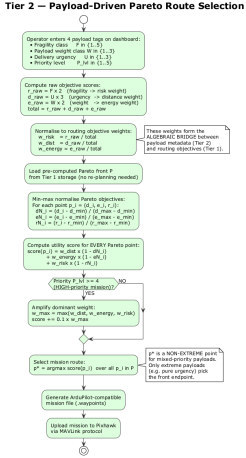
# Implementation Status

| Component | Status | Evidence / Remarks |
|---|---|---|
| Literature review & problem formulation | Completed | 3 papers surveyed; 3 gaps formalised; real-world evidence added |
| System design & architecture | Completed | Two-tier architecture, NAMOA* algorithm, and algebraic bridge designed; diagrams produced |
| Airspace graph construction | In Progress | Actively building the airspace graph model and 3-objective edge-cost assignment for the target delivery region |
| Tier 1: Multi-Objective A* (NAMOA*) engine | Planned | NAMOA* with 3-objective cost vectors + label-dominance pruning specified; implementation scheduled for Phase II |
| Tier 2: Payload-driven selector | Planned | Algebraic bridge and utility-scoring $\arg\max$ selection formalised; coding scheduled for Phase II |
| Benchmarking & evaluation | Planned | Metrics defined (HV, $\Delta$, IGD, selection accuracy, runtime); benchmark harness vs. single-obj. A* and MOPSO planned for Phase II |

# Prototype / Implementation Diagrams




Tier 1: Multi-Objective A* (NAMOA*) Pareto Front Construction




Tier 2: Payload-Driven Pareto Route Selection

# Expected Results and Performance Targets

## Payload Preset Weight Derivation

| Preset | $w_{\text{risk}}$ | $w_{\text{dist}}$ | $w_{\text{energy}}$ |
|---|---|---|---|
| Fragile medical ($F$=5, $W$=1, $U$=1) | 0.667 | 0.200 | 0.133 |
| Emergency medicine ($F$=1, $W$=1, $U$=3) | 0.144 | 0.712 | 0.144 |
| Disaster relief ($F$=1, $W$=3, $U$=1) | 0.182 | 0.273 | 0.545 |
| Mixed cargo ($F$=3, $W$=2, $U$=2) | 0.375 | 0.375 | 0.250 |

Only emergency medicine selects an extreme Pareto point. **All other presets select non-extreme interior points** no single-objective baseline can produce.

## Performance Targets

- HV: $\geq$20% higher than single-obj. A*
- Pareto front: 5–20 non-dominated routes per scenario
- Tier 2 selection latency: $<$50 ms on RPi 5
- Tier 2 accuracy: correct priority objective ranked highest across all 4 presets
- Hardware: ArduPilot SITL + outdoor delivery flight

# Testing and Validation Plan

| Test ID | Test Scenario | Expected Result | Status |
|---|---|---|---|
| TC-01 | Valid payload tags ($F, W, U, P_{\text{lvl}}$) input | Correct $w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}}$ derived; weights sum to 1.0 | Planned |
| TC-02 | NAMOA* on 50/200/500-node graphs (S1/S2/S3) | Pareto front with zero dominated routes; HV $>$ single-obj. A* | Planned |
| TC-03 | High fragility ($F$=5) vs. emergency medicine ($U$=3) | Significantly different non-extreme routes selected from same front | Planned |
| TC-04 | Disconnected graph (no feasible path) | Graceful error with clear user message | Planned |
| TC-05 | RPi 5 MAVLink mission upload to Pixhawk via SITL | Drone accepts mission and executes flight in ArduPilot SITL | Planned |

## Validation Method

Unit tests (pytest), MOPSO benchmarking (30 runs, swarm = 100, 200 iterations), ArduPilot SITL integration tests, and outdoor UAV flight tests with GPS telemetry logs.

# Project Timeline (Jul 2026 – Mar 2027)

| Activity | Jul | Aug | Sep | Oct | Nov | Dec | Jan | Feb | Mar |
|---|---|---|---|---|---|---|---|---|---|
| Literature survey & formulation | | | | | | | | | |
| System design & architecture | | | | | | | | | |
| Tier 1: Multi-Objective A* | | | | | | | | | |
| Tier 2: Payload-Driven Selector | | | | | | | | | |
| Software simulation & benchmarking | | | | | | | | | |
| Hardware integration (RPi 5 + Pixhawk) | | | | | | | | | |
| Outdoor flight tests | | | | | | | | | |
| Documentation & report writing | | | | | | | | | |



**Review I:** August 2026 **Final Submission:** March 2027

# Team Responsibilities

| **Team Member** | Responsibilities | Current Status |
|---|---|---|
| **Thirumurugan RA** | System architecture design, Tier 1 NAMOA* engine implementation, airspace graph construction (OSMnx), edge cost vectorization | In Progress |
| **Meghna Manimaran** | Algebraic bridge design, Tier 2 payload selector implementation, benchmarking suite development, documentation | In Progress |
| **Shrika SP Thota** | Literature survey & problem formulation, risk-cost model design, benchmarking methodology, Pareto-front visualization and result analysis | In Progress |

# Challenges, Risks, and Mitigation

| Risk / Challenge | Possible Impact | Mitigation |
|---|---|---|
| Graph complexity (large OSM urban network) | Slow Pareto front construction | Spatial pruning, NAMOA* node-level label-dominance check, epsilon-dominance to limit front size |
| Hardware failure (RPi / Pixhawk) | Flight tests delayed | ArduPilot SITL validation first; spare hardware budget |
| Pareto front size too large | High memory and selection latency | Apply epsilon-pruning; front bounded to $\leq$100 routes |
| MAVLink integration issues | Mission upload failures | Validate protocol end-to-end in SITL before real drone flights |

# Conclusion and Future Work

## Conclusion

This project addresses the critical gap of payload-blind UAV routing by proposing a two-tier payload-aware pipeline. Tier 1 builds a Pareto-optimal route set via NAMOA*; Tier 2 selects the mission route using an algebraic bridge that maps fragility, urgency, and payload weight to routing objective weights. Utility scoring over the full Pareto front replaces context-blind diversity criteria, enabling non-extreme route selection for mixed payloads that no single-objective baseline can produce.

## Future Work

- Outdoor flight validation on a real RPi 5 + Pixhawk drone in a simulated urban delivery environment.
- Extend to multi-drone coordination with shared Pareto fronts.
- Incorporate real-time weather and dynamic obstacle data for in-flight re-routing with adaptive front refresh thresholds.

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# Thank You

## Questions and Discussion

Thirumurugan RA · Meghna Manimaran · Shrika SP Thota

Guide: Sujaudeen N, Assistant Professor